The Inequalities Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: SOLVE, COMPOUND, REASON and APPLY (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** (SOLVE)
- ☐ **A2** (SOLVE)
- ☐ **A3** (SOLVE)
- ☐ **A4** (REASON)
- ☐ **A5** (COMPOUND)
- ☐ **A6** (APPLY)

Entrées

10 pts each · choose 4

- ☐ **E1** (SOLVE)
- ☐ **E2** (SOLVE)
- ☐ **E3** (COMPOUND)
- ☐ **E4** (COMPOUND)
- ☐ **E5** (APPLY)
- ☐ **E6** (REASON)

Chef's Specials

20 pts each · choose 2

- ☐ **S1** (APPLY)
- ☐ **S2** (COMPOUND)
- ☐ **S3** (REASON)
- ☐ **S4** (APPLY)

My skill check (circle each skill you have covered): SOLVE · COMPOUND · REASON · APPLY

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 SOLVE ☐ My pick · 5 pts

Solve: $x + 8 > 3$

A2 SOLVE ☐ My pick · 5 pts

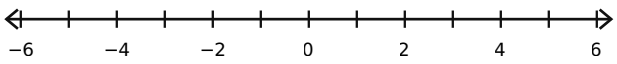
Solve: $-3x \le 12$

A3 SOLVE ☐ My pick · 5 pts

Solve: $2x - 5 < 9$

A4 REASON ☐ My pick · 5 pts

Graph the solution of $x \ge -2$ on the number line.



A5 COMPOUND ☐ My pick · 5 pts

Solve: $-3 < x - 2 \le 4$

A6 APPLY ☐ My pick · 5 pts

Movie tickets cost $9 each. Write an inequality for the number of tickets t Sam can buy with $50, solve it, and say what the answer means.

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 SOLVE ☐ My pick · 10 pts

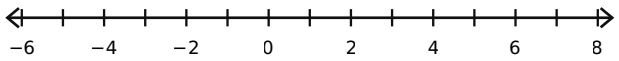
Solve: $5 - 2(x + 1) \geq 3x + 8$

E2 SOLVE ☐ My pick · 10 pts

Solve: $\dfrac{x - 4}{-3} < 2$

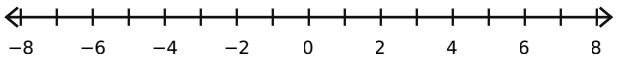
E3 COMPOUND ☐ My pick · 10 pts

Solve and graph: $-5 \leq 3x + 4 < 19$



E4 COMPOUND ☐ My pick · 10 pts

Solve and graph: $2x - 1 \leq -7$ **or** $4x + 3 > 15$



E5 APPLY ☐ My pick · 10 pts

A gym charges a $25 joining fee plus $12 per month. Maya can spend at most $145. Write and solve an inequality for the number of months m she can belong, and state the answer in words.

E6 REASON ☐ My pick · 10 pts

Ravi solved $-2x + 6 > 14$ like this:

$-2x > 8$

$x > -4$

Find his mistake, explain it, and give the correct solution.

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

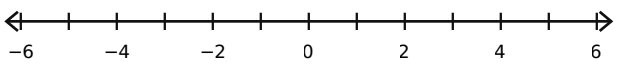
Name: _____

S1 APPLY ☐ My pick · 20 pts

A taxi charges $4 plus $3 per mile. Jordan wants to spend **at least $13 and at most $28**. Write a compound inequality for the number of miles m, solve it, and describe the trips Jordan can take.

S2 COMPOUND ☐ My pick · 20 pts

Solve and graph: $-4 < \frac{5 - 3x}{2} \leq 7$

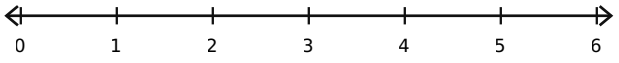


S3 REASON ☐ My pick · 20 pts

The solution of $2x + a > 11$ is $x > 4$.

(a) Find the value of a and show how you know.

(b) Using that value of a, solve $-2x + a > 11$.

S4 APPLY ☐ My pick · 20 pts

A greenhouse is 60°F when its heater turns on, and it warms 5°F every hour. Plants grow best when the temperature is **at least 65°F but below 80°F**. Write and solve a compound inequality for the number of hours h the heater has been on, then graph the solution.